

UNIVERSIDAD TECNOLÓGICA NACIONAL
FACULTAD REGIONAL ROSARIO
PRÁCTICA III: PL MÉTODO SIMPLEX

INVESTIGACIÓN OPERATIVA

Ejercicio1

$$\text{Max } Z = 2 X_1 + 3 X_2 + 0 X_3 + 0 X_4 + 0 X_5$$

S.a.::

$$4 X_1 + 4 X_2 + 1 X_3 = 320$$

$$2 X_1 + 4 X_2 + 1 X_4 = 240$$

$$8 X_1 + 4 X_2 + 1 X_5 = 560$$

$$X_j \geq 0$$

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
0	A_3	4	4	1	0	0	320	80
0	A_4	2	4	0	1	0	240	60
0	A_5	8	4	0	0	1	560	560/4
Z_j		0	0					
$C_j - Z_j$		2	3				$Z = 0$	

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
0	A_3	2	0	1	-1	0	80	40
3	A_2	1/2	1	0	1/4	0	60	120
0	A_5	6	0	0	-1	1	320	320/6
Z_j		3/2			3/4			
$C_j - Z_j$		1/2			-3/4		$Z = 180$	

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
2	A_1	1	0	1/2	-1/2	0	40	
3	A_2	0	1	-1/4	1/2	0	40	
0	A_5	0	0	-3	2	1	80	
Z_j				1/4	1/2			
$C_j - Z_j$				-1/4	-1/2		$Z^* = 200$	

$$X^{*T} = (40, 40, 0, 0, 80)$$

$$Z^* = 200$$

Ejercicio 2

a)

$$\text{Max } z = 3x_1 + 9x_2 + 0x_3 + 0x_4$$

s. a

$$1x_1 + 4x_2 + 1x_3 = 8$$

$$1x_1 + 2x_2 + 1x_4 = 4$$

$$x_j \geq 0 \quad j = 1; 2; \dots; 4$$

			3	9	0	0			x_i/y_{ij} ($y_{ij} > 0$)	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i		
	0	A_3	1	4	1	0		8	2	
	0	A_4	1	2	0	1		4	2	
	z_j		0	0				$z = 0$		
	$c_i - z_j$		3	9						

			3	9	0	0			x_i/y_{ij} ($y_{ij} > 0$)	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i		
	9	A_2	1/4	1	1/4	0		2	8	
	0	A_4	1/2	0	-1/2	1		0	0	
	z_j		9/4		9/4			$z = 18$		
	$c_i - z_j$		3/4		-9/4					

			3	9	0	0			x_i/y_{ij} ($y_{ij} > 0$)	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i		
	9	A_2	0	1	1/2	-1/2		2		
	3	A_1	1	0	-1	2		0		
	z_j				3/2	3/2		$z = 18$		
	$c_i - z_j$				-3/2	-3/2				

$$x_1^* = 0; x_2^* = 2; x_3^* = 0; x_4^* = 0; z^* = 18$$

Solución degenerada

b) $\text{Min } W = 5 X_1 + 6 X_2 - 7 X_3$
S.a.:

$$\begin{aligned} X_1 + 5 X_2 - 3 X_3 &\geq 15 \\ 5 X_1 - 6 X_2 + 10 X_3 &\leq 20 \\ X_1 + X_2 + X_3 &= 5 \end{aligned}$$

$$X_j \geq 0$$

$$\text{Min } W = 5 X_1 + 6 X_2 - 7 X_3 \quad \text{ó} \quad \text{Max } Z = -5 X_1 - 6 X_2 + 7 X_3$$

s.a

$$\begin{aligned} X_1 + 5 X_2 - 3 X_3 - 1 X_4 + 0 X_5 + 1 X_6 + 0 X_7 &= 15 \\ 5 X_1 - 6 X_2 + 10 X_3 + 0 X_4 + 1 X_5 + 0 X_6 + 0 X_7 &= 20 \\ X_1 + X_2 + X_3 + 0 X_4 + 0 X_5 + 0 X_6 + 1 X_7 &= 5 \\ X_j &\geq 0 \end{aligned}$$

Siendo:

$$\text{Max } Z = -5 X_1 - 6 X_2 + 7 X_3 + 0 X_4 + 0 X_5 - M X_6 - M X_7$$

	C_j	-5	-6	7	0	0	-M	-M		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-M	A6	1	5	-3	-1	0	1	0	15	3
0	A5	5	-6	10	0	1	0	0	20	-
-M	A7	1	1	1	0	0	0	1	5	5
Z_j		-2M	-6M	2M	M					
$C_j - Z_j$		-5+2M	-6+6M	7-2M	-M				$Z = -20M$	



	C_j	-5	-6	7	0	0	-M	-M		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-6	A2	1/5	1	-3/5	-1/5	0	1/5	0	3	-
0	A5	31/5	0	32/5	-6/5	1	6/5	0	38	95/16
-M	A7	4/5	0	8/5	1/5	0	-1/5	1	2	5/4
Z_j		-6/5-4/5M		18/5-8/5M	6/5-M/5		-6/5+ M/5			
$C_j - Z_j$		-19/5+4/5M		17/5+8/5M	-6/5+M/5		6/5-6M/5		$Z = -18-2M$	



	C_j	-5	-6	7	0	0	-M	-M		
C_i	A_i A_j	A₁	A₂	A₃	A₄	A₅	A₆	A₇	X_i	X_i / Y_{ij}
-6	A₂	$\frac{1}{2}$	1	0	-1/8	0	1/8	3/8	15/4	
0	A₅	3	0	0	-2	1	2	-4	30	
7	A₃	$\frac{1}{2}$	0	1	1/8	0	-1/8	5/8	5/4	
Z_j		1/2			13/8		-13/8	17/8		
C_j - Z_j		-11/2			-13/8		-M+13/8	-M-17/8	Z* = -55/4	W* = 55/4

$$X^{*T} = (0, 15/4, 5/4, 0, 30)$$

$$W^* = 55/4$$

b) Resolver utilizando el Método de las dos Fases

FASE I

Programa auxiliar:

$$\text{Min } \Phi = X_6 + X_7 \quad \text{o} \quad \text{Max } \zeta = -X_6 - X_7$$

s. a

$$X_1 + 5X_2 - 3X_3 - 1X_4 + 0X_5 + 1X_6 + 0X_7 = 15$$

$$5X_1 - 6X_2 + 10X_3 + 0X_4 + 1X_5 + 0X_6 + 0X_7 = 20$$

$$X_1 + X_2 + X_3 + 0X_4 + 0X_5 + 0X_6 + 1X_7 = 5$$

$$x_j \geq 0$$

	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-1	A6	1	5	-3	-1	0	1	0	15	3
0	A5	5	-6	10	0	1	0	0	20	-
-1	A7	1	1	1	0	0	0	1	5	5
Z_j		-2	-6	2	1					
$C_j - Z_j$		2	6	-2	-1				$\zeta = -20$	



	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
0	A2	1/5	1	-3/5	-1/5	0	1/5	0	3	-
0	A5	31/5	0	32/5	-6/5	1	6/5	0	38	95/16
-1	A7	4/5	0	8/5	1/5	0	-1/5	1	2	5/4
Z_j		-4/5		-8/5	-1/5		1/5			
$C_j - Z_j$		4/5		8/5	1/5		-6/5		$\zeta = -2$	



	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
0	A2	$\frac{1}{2}$	1	0	$-\frac{1}{8}$	0	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{15}{4}$	
0	A5	3	0	0	-2	1	2	-4	30	
0	A3	$\frac{1}{2}$	0	1	$\frac{1}{8}$	0	$-\frac{1}{8}$	$\frac{5}{8}$	$\frac{5}{4}$	
Z_j		0			0		0	0		
$C_j - Z_j$		0			0		-1	-1	$\zeta^* = 0$	

FASE II

PL Original: $\text{Max } Z = -5 X_1 - 6 X_2 + 7 X_3$

	C_j	-5	-6	7	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-6	A2	$\frac{1}{2}$	1	0	$-\frac{1}{8}$	0			$\frac{15}{4}$	
0	A5	3	0	0	-2	1			30	
7	A3	$\frac{1}{2}$	0	1	$\frac{1}{8}$	0			$\frac{5}{4}$	
Z_j		$\frac{1}{2}$			$\frac{13}{8}$					
$C_j - Z_j$		$-\frac{11}{2}$			$-\frac{13}{8}$				$Z^* = -\frac{55}{4}$ $W^* = \frac{55}{4}$	

$$X^{*T} = (0, \frac{15}{4}, \frac{5}{4}, 0, 30)$$

$$W^* = \frac{55}{4}$$

c) $\text{Max } Z = 3 X_1 + 2 X_2$
S.a.:
 $2 X_1 + X_2 \leq 2$
 $3 X_1 + 4 X_2 \geq 12$
 $X_j \geq 0$

Llevarlo a la forma estándar

$\text{Máx } Z = 3 X_1 + 2 X_2 + 0 X_3 + 0 X_4$
S.a.:
 $2 X_1 + X_2 + 1 X_3 + 0 X_4 = 2$
 $3 X_1 + 4 X_2 + 0 X_3 - 1 X_4 + X_5 = 12$
 $X_j \geq 0$

b) Resolverlo por tablas (Método de las dos fases)

FASE I

Programa auxiliar:

$\text{Max } \zeta = -X_5$
S.a.:
 $2 X_1 + X_2 + 1 X_3 + 0 X_4 = 2$
 $3 X_1 + 4 X_2 + 0 X_3 - 1 X_4 + X_5 = 12$
 $X_j \geq 0$

	C_j	0	0	0	0	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
0	A3	2	1	1	0	0	2	2
-1	A5	3	4	0	-1	1	12	3
Z_j		-3	-4		1			
$C_j - Z_j$		3	4		-1		$\zeta = -12$	



	C_j	0	0	0	0	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
0	A2	2	1	1	0	0	2	
-1	A5	-5	0	-4	-1	1	4	
Z_j		5		4	1			
$C_j - Z_j$		-5		-4	-1		$\zeta = -4$	

Se alcanzó la condición de **óptimo con $\zeta \neq 0$**
y la variable ficticia: **$X_5 = 4 \neq 0$**

Luego, el PL original **NO tiene Solución Factible**

d)

$$\text{Max } Z = 8x_1 + 30x_2$$

s.a

$$10x_1 + 12x_2 - 1x_3 + 0x_4 + 0x_5 + 1x_6 = 60$$

$$3x_1 - 4x_2 + 0x_3 + 1x_4 + 0x_5 + 0x_6 = 12$$

$$-3x_1 + 2x_2 + 0x_3 + 0x_4 + 1x_5 + 0x_6 = 30$$

$$x_j \geq 0$$

	C_j	8	30	0	0	0	-M		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i / Y_{ij}
-M	A6	10	12	-1	0	0	1	60	5
0	A4	3	-4	0	1	0	0	12	-
0	A5	-3	2	0	0	1	0	30	15
Z_j		-10M	-12M	M					
$C_j - Z_j$		8+10M	30+12M	-M				Z = -60M	



	C_j	8	30	0	0	0	-M		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	A_5	A6	X_i	X_i / Y_{ij}
30	A2	5/6	1	-1/12	0	0	1/12	5	-
0	A4	19/3	0	-1/3	1	0	1/3	32	-
0	A5	-14/3	0	1/6	0	1	1/6	20	120
Z_j		25		-15/6					
$C_j - Z_j$		-17		15/6				Z = 150	



	C_j	8	30	0	0	0		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i / Y_{ij}
30	A2	-3/2	1	0	0	1/2	15	-
0	A4	-3	0	0	1	2	72	-
0	A3	-28	0	1	0	6	120	-
Z_j		-45				15		
$C_j - Z_j$		53				-15	Z = 450	



$$Z = \infty$$

b) Resolver utilizando el Método de las dos Fases

FASE I

Programa auxiliar:

$$\text{Min } \Phi = X_6 + X_7 \quad \text{o} \quad \text{Max } \zeta = -X_6 - X_7$$

s. a

$$X_1 + 5X_2 - 3X_3 - 1X_4 + 0X_5 + 1X_6 + 0X_7 = 15$$

$$5X_1 - 6X_2 + 10X_3 + 0X_4 + 1X_5 + 0X_6 + 0X_7 = 20$$

$$X_1 + X_2 + X_3 + 0X_4 + 0X_5 + 0X_6 + 1X_7 = 5$$

$$x_j \geq 0$$

	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-1	A6	1	5	-3	-1	0	1	0	15	3
0	A5	5	-6	10	0	1	0	0	20	-
-1	A7	1	1	1	0	0	0	1	5	5
Z_j		-2	-6	2	1					
$C_j - Z_j$		2	6	-2	-1				$\zeta = -20$	



	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
0	A2	1/5	1	-3/5	-1/5	0	1/5	0	3	-
0	A5	31/5	0	32/5	-6/5	1	6/5	0	38	95/16
-1	A7	4/5	0	8/5	1/5	0	-1/5	1	2	5/4
Z_j		-4/5		-8/5	-1/5		1/5			
$C_j - Z_j$		4/5		8/5	1/5		-6/5		$\zeta = -2$	



e) $\text{Min } W = 24 X_1 + 36 X_2 + 24 X_3 + 36 X_4$

S.a

$$6 X_1 + 4 X_2 = 100$$

$$2 X_3 + 3 X_4 = 24$$

$$X_j \geq 0$$

Por penalización

$$\text{Min } W = 24 X_1 + 36 X_2 + 24 X_3 + 36 X_4 + M X_5 + M X_6$$

S.a:

$$6 X_1 + 4 X_2 + 0 X_3 + 0 X_4 + 1 X_5 + 0 X_6 = 100$$

$$0 X_1 + 0 X_2 + 2 X_3 + 3 X_4 + 0 X_5 + 1 X_6 = 24$$

$$X_j \geq 0$$

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i / Y_{ij}
M	A_5	6	4	0	0	1	0	100	100/6
M	A_6	0	0	2	3	0	1	24	-
Z_j		6M	4M	2M	3M				
$C_j - Z_j$		-6M+24	-4M+36	-2M+24	-3M+36			$W = 124M$	



	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i / Y_{ij}
24	A_1	1	4/6	0	0	1/6	0	100/6	-
M	A_6	0	0	2	3	0	1	24	8
Z_j			16	2M	3M	4			
$C_j - Z_j$			20	-2M+24	-3M+36	-4+M		$W = 400+24M$	



	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i / Y_{ij}
24	A1	1	4/6	0	0	1/6	0	100/6	
36	A4	0	0	2/3	1	0	1/3	8	
Z_j			16	24		4	12		
$C_j - Z_j$			20	0		-4+M	-12+M	$W^* =$	688

$$X^{*T} = (100/6, 0, 0, 8)$$

$$W^* = 688$$

Solución alternativa:

Como $C_3 - Z_3 = 0$ siendo X_3 NO básica, la solución alternativa se obtiene haciendo ingresar a A_3 a la base:

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i / Y_{ij}
24	A1	1	4/6	0	0	1/6	0	100/6	
24	A3	0	0	1	3/2	0	1/2	12	
Z_j			16		36	4	12		
$C_j - Z_j$			-20		0	-4+M	-12+M	$W^* =$	688

$$X^{*T} = (100/6, 0, 12, 0)$$

$$W^* = 688$$

ALGORITMO MATRICIAL

Ejercicio 1

1. Se busca una solución básica inicial.

$$\text{Max } z = 2x_1 + 3x_2 + 0x_3 + 0x_4 + 0x_5$$

s. a

$$4x_1 + 4x_2 + x_3 = 320$$

$$2x_1 + 4x_2 + x_4 = 240$$

$$8x_1 + 4x_2 + x_5 = 560$$

$$x_j \geq 0 \quad j = 1; 2; \dots; 5$$

$$B = (A_3, A_4, A_5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}; N = (A_1, A_2) = \begin{pmatrix} 4 & 4 \\ 2 & 4 \\ 8 & 4 \end{pmatrix}$$

$$x_B = (x_3, x_4, x_5)^T = (320; 240; 560)^T; x_N = (x_1, x_2)^T = (0; 0)^T$$

$$z = c_B x_B = (0; 0; 0)(320; 240; 560)^T = 0$$

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 4 & 4 \\ 2 & 4 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} y_{31} & y_{32} \\ y_{41} & y_{42} \\ y_{51} & y_{52} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_1, z_2) = c_B Y = (0; 0; 0) \begin{pmatrix} 4 & 4 \\ 2 & 4 \\ 8 & 4 \end{pmatrix} = (0; 0)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = 2 - 0 = 2$$

$$c_2 - z_2 = 3 - 0 = 3 \Rightarrow \text{entra } A_2$$

5. Se determina el vector que sale.

$$\theta_2 = \min \left\{ \frac{x_3}{y_{32}}, \frac{x_4}{y_{42}}, \frac{x_5}{y_{52}} \right\} = \min \left\{ \frac{320}{4}, \frac{240}{4}, \frac{560}{4} \right\} = 60 \Rightarrow \text{sale } A_4$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = x_1 = 0; x'_2 = \theta_2 = 60; x'_3 = x_3 - \theta_2 y_{32} = 320 - 60 \cdot 4 = 80$$

$$x'_4 = x_4 - \theta_2 y_{42} = 240 - 60 \cdot 4 = 0; x'_5 = x_5 - \theta_2 y_{52} = 560 - 60 \cdot 4 = 320$$

$$x_B = (x_2, x_3, x_5)^T = (60; 80; 320)^T; x_N = (x_1, x_4)^T = (0; 0)^T$$

$$z = c_B x_B = (3; 0; 0)(60; 80; 320)^T = 180$$

$$B = (A_2, A_3, A_5) = \begin{pmatrix} 4 & 1 & 0 \\ 4 & 0 & 0 \\ 4 & 0 & 1 \end{pmatrix}; N = (A_1, A_4) = \begin{pmatrix} 4 & 0 \\ 2 & 1 \\ 8 & 0 \end{pmatrix}$$

Primera iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 0 & 1/4 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 0 & 1/4 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 2 & 1 \\ 8 & 0 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/4 \\ 2 & -1 \\ 6 & -1 \end{pmatrix} = \begin{pmatrix} y_{21} & y_{24} \\ y_{31} & y_{34} \\ y_{51} & y_{54} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_1, z_4) = c_N Y = (3; 0; 0) \begin{pmatrix} 1/2 & 1/4 \\ 2 & -1 \\ 6 & -1 \end{pmatrix} = (3/2; 3/4)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = 2 - 3/2 = 1/2 \Rightarrow \text{entra } A_1$$

$$c_4 - z_4 = 0 - 3/4 = -3/4$$

5. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_2}{y_{21}}, \frac{x_3}{y_{31}}, \frac{x_5}{y_{51}} \right\} = \min \left\{ \frac{60}{1/2}, \frac{80}{2}, \frac{320}{6} \right\} = 40 \Rightarrow \text{sale } A_3$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = \theta_1 = 40; x'_2 = x_2 - \theta_1 y_{21} = 60 - 40 \cdot 1/2 = 40; x'_3 = x_3 - \theta_1 y_{31} = 80 - 40 \cdot 2 = 0$$

$$x'_4 = 0; x'_5 = x_5 - \theta_1 y_{51} = 320 - 40 \cdot 6 = 80$$

$$x_B = (x_1, x_2, x_5)^T = (40; 40; 80)^T; x_N = (x_3, x_4)^T = (0; 0)^T$$

$$z = c_B x_B = (2; 3; 0)(40; 40; 80)^T = 200$$

$$B = (A_1, A_2, A_5) = \begin{pmatrix} 4 & 4 & 0 \\ 2 & 4 & 0 \\ 8 & 4 & 1 \end{pmatrix}; N = (A_3, A_4) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Segunda iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/4 & 1/2 & 0 \\ -3 & 2 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/4 & 1/2 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ -1/4 & 1/2 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} y_{13} & y_{14} \\ y_{23} & y_{24} \\ y_{53} & y_{54} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_3, z_4) = c_B Y = (2; 3; 0) \begin{pmatrix} 1/2 & -1/2 \\ -1/4 & 1/2 \\ -3 & 2 \end{pmatrix} = (1/4; 1/2)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_3 - z_3 = 0 - 1/4 = -1/4$$

$$c_4 - z_4 = 0 - 1/2 = -1/2$$

Se verifica la condición de optimización,

$$\boxed{x_1^* = 40; x_2^* = 40; x_3^* = 0; x_4^* = 0; x_5^* = 80; z^* = 200}$$

Ejercicio 2

1. Se busca una solución básica inicial.

$$\text{Max } z = 3x_1 + 9x_2 + 0x_3 + 0x_4$$

s. a

$$1x_1 + 4x_2 + 1x_3 = 8$$

$$1x_1 + 2x_2 + 1x_4 = 4$$

$$x_j \geq 0 \quad j = 1; 2; \dots; 4$$

$$B = (A_3, A_4) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; N = (A_1, A_2) = \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix}$$

$$x_B = (x_3, x_4)^T = (8; 4)^T; x_N = (x_1, x_2)^T = (0; 0)^T$$

$$z = c_B x_B = (0; 0)(8; 4)^T = 0$$

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} y_{31} & y_{32} \\ y_{41} & y_{42} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_1, z_2) = c_N Y = (0; 0) \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix} = (0; 0)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = 3 - 0 = 3$$

$$c_2 - z_2 = 9 - 0 = 9 \Rightarrow \text{entra } A_2$$

5. Se determina el vector que sale.

$$\theta_2 = \min \left\{ \frac{x_3}{y_{32}}, \frac{x_4}{y_{42}} \right\} = \min \left\{ \frac{8}{4}, \frac{4}{2} \right\} = 2 \Rightarrow \underline{\text{sale } A_3} \text{ o } A_4$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = 0; x'_2 = \theta_2 = 2; x'_3 = x_3 - \theta_2 y_{32} = 8 - 2 \cdot 4 = 0$$

$$x'_4 = x_4 - \theta_2 y_{42} = 4 - 2 \cdot 2 = 0$$

$$x_B = (x_2, x_4)^T = (2; 0)^T; x_N = (x_1, x_3)^T = (0; 0)^T$$

$$z = c_B x_B = (9; 0)(2; 0)^T = 18$$

$$B = (A_2, A_4) = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}; N = (A_1, A_3) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Primera iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1/4 & 0 \\ -1/2 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1/4 & 0 \\ -1/2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1/4 & 1/4 \\ 1/2 & -1/2 \end{pmatrix} = \begin{pmatrix} y_{21} & y_{23} \\ y_{41} & y_{43} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_1, z_3) = c_B Y = (9; 0) \begin{pmatrix} 1/4 & 1/4 \\ 1/2 & -1/2 \end{pmatrix} = (9/4; 9/4)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = 3 - 9/4 = 3/4 \Rightarrow \text{entra } A_1$$

$$c_3 - z_3 = 0 - 9/4 = -9/4$$

5. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_2}{y_{21}}, \frac{x_4}{y_{41}} \right\} = \min \left\{ \frac{2}{1/4}; \frac{0}{1/2} \right\} = 0 \Rightarrow \text{sale } A_4$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = \theta_1 = 0; x'_2 = x_2 - \theta_1 y_{21} = 2 - 0 \cdot 1/4 = 2; x'_3 = 0$$

$$x'_4 = 0;$$

$$x_B = (x_2, x_1)^T = (2; 0)^T; x_N = (x_3, x_4)^T = (0; 0)^T$$

$$z = c_B x_B = (9; 3)(2; 0)^T = 18$$

$$B = (A_2, A_1) = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix}; N = (A_3, A_4) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Segunda iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ -1 & 2 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1/2 & -1/2 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} y_{B3} & y_{B4} \\ y_{N3} & y_{N4} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_3, z_4) = c_B Y = (9; 3) \begin{pmatrix} 1/2 & -1/2 \\ -1 & 2 \end{pmatrix} = (3/2; 3/2)$$

2. Se calculan las diferencias $c_j - z_j$.

$$c_3 - z_3 = 0 - 3/2 = -3/2$$

$$c_4 - z_4 = 0 - 3/2 = -3/2$$

Se verifica la condición de optimización,

$$\boxed{x_1^* = 0; x_2^* = 2; x_3^* = 0; x_4^* = 0; z^* = 18}$$

Solución óptima degenerada

Ejercicio 3

3.a)

1. Se busca una solución básica inicial.

$$\text{Min } w = -x_1 - x_2 + 0x_3 + 0x_4 + 0x_5$$

s. a

$$\begin{aligned} -x_1 + x_2 + x_3 &= 2 \\ x_1 + 2x_2 + x_4 &= 6 \\ 2x_1 + x_2 + x_5 &= 6 \\ x_j &\geq 0 \quad j = 1; 2; \dots; 5 \end{aligned}$$

$$B = (A_3, A_4, A_5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}; N = (A_1, A_2) = \begin{pmatrix} -1 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$x_B = (x_3, x_4, x_5)^T = (2; 6; 6)^T; x_N = (x_1, x_2)^T = (0; 0)^T$$

$$w = c_B x_B = (0; 0; 0)(2; 6; 6)^T = 0$$

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} -1 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} y_{31} & y_{32} \\ y_{41} & y_{42} \\ y_{51} & y_{52} \end{pmatrix}$$

3. Se determinan los z_j .

$$z_1 = \sum_{i \in I_B} c_i y_{i1} = c_3 y_{31} + c_4 y_{41} + c_5 y_{51} = 0$$

$$z_2 = \sum_{i \in I_B} c_i y_{i2} = c_3 y_{32} + c_4 y_{42} + c_5 y_{52} = 0 \quad \text{o bien, } (z_1, z_2) = c_B Y = (0; 0; 0) \begin{pmatrix} -1 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix} = (0; 0)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = -1 - 0 = -1 \Rightarrow \text{entra } A_1$$

$$c_2 - z_2 = -1 - 0 = -1$$

5. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_i}{y_{i1}}, \text{ con } y_{i1} > 0 \right\} = \min \left\{ \frac{x_3}{y_{31}}, \frac{x_4}{y_{41}}, \frac{x_5}{y_{51}} \right\} = \min \left\{ -; \frac{6}{1}; \frac{6}{2} \right\} = 3 \Rightarrow \text{sale } A_5 (y_{31} < 0)$$

6. Se calcula la nueva solución, el w asociado y se vuelve al punto 2.

$$x'_1 = \theta_1 = 3; x'_2 = x_2 = 0; x'_3 = x_3 - \theta_1 y_{31} = 2 - 3(-1) = 5$$

$$x'_4 = x_4 - \theta_1 y_{41} = 6 - 3 \cdot 1 = 3; x'_5 = x_5 - \theta_1 y_{51} = 6 - 3 \cdot 2 = 0$$

$$x_B = (x_1, x_3, x_4)^T = (3; 5; 3)^T; x_N = (x_2, x_5)^T = (0; 0)^T$$

$$w = c_B x_B = (-1; 0; 0)(3; 5; 3)^T = -3$$

$$B = (A_1, A_3, A_4) = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix}; N = (A_2, A_5) = \begin{pmatrix} 1 & 0 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$$

Primera iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 1/2 \\ 0 & 1 & -1/2 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 1/2 \\ 0 & 1 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 3/2 & 1/2 \\ 3/2 & -1/2 \end{pmatrix} = \begin{pmatrix} y_{12} & y_{15} \\ y_{32} & y_{35} \\ y_{42} & y_{45} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_2, z_5) = c_B Y = (-1; 0; 0) \begin{pmatrix} 1/2 & 1/2 \\ 3/2 & 1/2 \\ 3/2 & -1/2 \end{pmatrix} = (-1/2; -1/2)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_2 - z_2 = -1 + 1/2 = -1/2 \Rightarrow \text{entra } A_2$$

$$c_5 - z_5 = 0 + 1/2 = 1/2$$

5. Se determina el vector que sale.

$$\theta_2 = \min \left\{ \frac{x_1}{y_{12}}, \frac{x_3}{y_{32}}, \frac{x_4}{y_{42}} \right\} = \min \left\{ \frac{3}{1/2}, \frac{5}{3/2}, \frac{3}{3/2} \right\} = 2 \Rightarrow \text{sale } A_4$$

6. Se calcula la nueva solución, el w asociado y se vuelve al punto 2.

$$x'_1 = x_1 - \theta_2 y_{12} = 3 - 2 \cdot 1/2 = 2; x'_2 = \theta_2 = 2; x'_3 = x_3 - \theta_2 y_{32} = 5 - 2 \cdot 3/2 = 2$$

$$x'_4 = x_4 - \theta_2 y_{42} = 3 - 2 \cdot 3/2 = 0; x'_5 = 0$$

$$x_B = (x_1, x_2, x_3)^T = (2; 2; 2)^T; x_N = (x_4, x_5)^T = (0; 0)^T$$

$$w = c_B x_B = (-1; -1; 0)(2; 2; 2)^T = -4$$

$$B = (A_1, A_2, A_3) = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 2 & 0 \\ 2 & 1 & 0 \end{pmatrix}; N = (A_4, A_5) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Segunda iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 0 & -1/3 & 2/3 \\ 0 & 2/3 & -1/3 \\ 1 & -1 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 0 & -1/3 & 2/3 \\ 0 & 2/3 & -1/3 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} y_{14} & y_{15} \\ y_{24} & y_{25} \\ y_{34} & y_{35} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_4, z_5) = c_B Y = (-1; -1; 0) \begin{pmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \\ -1 & 1 \end{pmatrix} = (-1/3; -1/3)$$

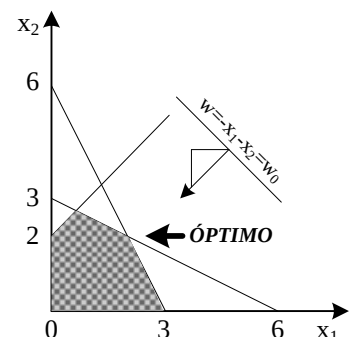
4. Se calculan las diferencias $c_j - z_j$.

$$c_4 - z_4 = 0 - (-1/3) = 1/3$$

$$c_5 - z_5 = 0 - (-1/3) = 1/3$$

Puesto que se verifica la condición de optimización, resulta

$$x_1^* = 2; x_2^* = 2; x_3^* = 2; x_4^* = 0; x_5^* = 0; w^* = -4$$



$$\text{Min } w = -3x_1 + 8x_2$$

s.a

$$4x_1 + x_2 \leq 13$$

$$2x_1 + 3x_2 \leq 6$$

$$x_j \geq 0$$

7. Se busca una solución básica inicial.

$$\text{Min } w = -3x_1 + 8x_2 + 0x_3 + 0x_4$$

s.a

$$4x_1 + x_2 + x_3 = 13$$

$$2x_1 + 3x_2 + x_4 = 6$$

$$x_j \geq 0 \quad j = 1; 2; \dots; 4$$

$$B = (A_3, A_4) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; N = (A_1, A_2) = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

$$x_B = (x_3, x_4)^T = (13; 6)^T; x_N = (x_1, x_2)^T = (0; 0)^T$$

$$w = c_B x_B = (0; 0)(13; 6)^T = 0$$

8. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} y_{31} & y_{32} \\ y_{41} & y_{42} \end{pmatrix}$$

9. Se determinan los z_j .

$$z_1 = \sum_{i \in I_B} c_i y_{i1} = c_3 y_{31} + c_4 y_{41} = 0$$

$$z_2 = \sum_{i \in I_B} c_i y_{i2} = c_3 y_{32} + c_4 y_{42} = 0 \quad \text{o bien, } (z_1, z_2) = c_B Y = (0; 0) \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} = (0; 0)$$

10. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = -3 - 0 = -3 \Rightarrow \text{entra } A_1$$

$$c_2 - z_2 = 8 - 0 = 8$$

11. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_i}{y_{i1}}, \text{ con } y_{i1} > 0 \right\} = \min \left\{ \frac{x_3}{y_{31}}, \frac{x_4}{y_{41}} \right\} = \min \left\{ \frac{13}{4}, \frac{6}{2} \right\} = 3 \Rightarrow \text{sale } A_4$$

12. Se calcula la nueva solución, el w asociado y se vuelve al punto 2.

$$x'_1 = \theta_1 = 3; x'_2 = x_2 = 0; x'_3 = x_3 - \theta_1 y_{31} = 13 - 3(4) = 1$$

$$x'_4 = x_4 - \theta_1 y_{41} = 6 - 3 \cdot 2 = 0$$

$$x_B = (x_1, x_3)^T = (3; 1)^T; x_N = (x_2, x_4)^T = (0; 0)^T$$

$$w = c_B x_B = (-3; 0)(3; 1)^T = -9$$

$$B = (A_1, A_3) = \begin{pmatrix} 4 & 1 \\ 2 & 0 \end{pmatrix}; N = (A_2, A_4) = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$$

Primera iteración

7. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 0 & 1/2 \\ 1 & -2 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 0 & 1/2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 3/2 & 1/2 \\ -5 & -2 \end{pmatrix} = \begin{pmatrix} y_{12} & y_{14} \\ y_{32} & y_{34} \end{pmatrix}$$

8. Se determinan los z_j .

$$(z_2, z_4) = c_B Y = (-3; 0) \begin{pmatrix} 3/2 & 1/2 \\ -5 & -2 \end{pmatrix} = (-9/2; -3/2)$$

9. Se calculan las diferencias $c_j - z_j$.

$$c_2 - z_2 = 8 - (-9/2) = 25/2$$

$$c_4 - z_4 = 0 - (-3/2) = 3/2$$

Puesto que se verifica la condición de optimización, resulta

$$\boxed{x_1^* = 3; x_2^* = 0; x_3^* = 1; x_4^* = 0; w^* = -9}$$

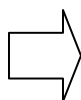
3) POR TABLAS Y MATRICIAL

$$\text{MAX } Z = 2X_1 + X_2$$

s.a.:

$$X_1 - X_2 \leq 10$$

$$2X_1 - X_2 \leq 40$$



$$\text{MAX } Z = 2X_1 + X_2 + 0X_3 + 0X_4$$

s.a.:

$$X_1 - X_2 + X_3 = 10$$

$$2X_1 - X_2 + X_4 = 40$$

$$x_j \geq 0$$

$$x_j \geq 0$$

	C_j	2	1	0	0		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	X_i	X_i / Y_{ij}
0	A_3	1	-1	1	0	10	10
0	A_4	2	-1	0	1	40	20
Z_j		0	0				
$C_j - Z_j$		2	1			$Z = 0$	

↑

	C_j	2	1	0	0		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	X_i	X_i / Y_{ij}
2	A_1	1	-1	1	0	10	-
0	A_4	0	1	-2	1	20	20
Z_j			-2	2			
$C_j - Z_j$			3	-2		$Z = 20$	

↑

	C_j	2	1	0	0		
C_i	$A_i \setminus A_j$	A_1	A_2	A_3	A_4	X_i	X_i / Y_{ij}
2	A_1	1	0	-1	1	30	-
1	A_2	0	1	-2	1	20	-
Z_j				-4	3		
$C_j - Z_j$				4	-3	$Z = \infty$	

↑

Aun no se ha llegado a la condición de óptimo pues $Z_3 - C_3 < 0$, sin embargo $y_{13} < 0$, por lo tanto A_1 no puede salir de la base y $y_{23} < 0$, por lo tanto A_2 tampoco puede salir de la base. La solución factible puede hacerse arbitrariamente grande. El conjunto de soluciones factibles es no acotado en la dirección de C .

1. Se busca una solución básica inicial.

$$\text{Max } z = 2x_1 + 1x_2 + 0x_3 + 0x_4$$

s. a

$$1x_1 - 1x_2 + 1x_3 = 10$$

$$2x_1 - 1x_2 + 1x_4 = 40$$

$$x_j \geq 0 \quad j = 1; 2; \dots; 4$$

$$B = (A_3, A_4) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; N = (A_1, A_2) = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$$

$$x_B = (x_3, x_4)^T = (10; 40)^T; x_N = (x_1, x_2)^T = (0; 0)^T$$

$$z = c_B x_B = (0; 0)(10; 40)^T = 0$$

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} y_{31} & y_{32} \\ y_{41} & y_{42} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_1, z_2) = c_B Y = (0; 0) \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} = (0; 0)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_1 - z_1 = 2 - 0 = 2 \Rightarrow \text{entra } A_1$$

$$c_2 - z_2 = 1 - 0 = 1$$

5. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_3}{y_{31}}, \frac{x_4}{y_{41}} \right\} = \min \left\{ \frac{10}{1}, \frac{40}{2} \right\} = 10 \Rightarrow \text{sale } A_3$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = \theta_1 = 10; x'_2 = 0; x'_3 = x_3 - \theta_1 y_{31} = 10 - 10 \cdot 1 = 0$$

$$x'_4 = x_4 - \theta_1 y_{41} = 40 - 10 \cdot 2 = 20$$

$$x_B = (x_1, x_4)^T = (10; 20)^T; x_N = (x_2, x_3)^T = (0; 0)^T$$

$$z = c_B x_B = (2; 0)(10; 20)^T = 20$$

$$B = (A_1, A_4) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}; N = (A_2, A_3) = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$$

Primera iteración

2. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} y_{12} & y_{13} \\ y_{22} & y_{23} \end{pmatrix}$$

3. Se determinan los z_j .

$$(z_2, z_3) = c_B Y = (2; 0) \begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix} = (-2; 2)$$

4. Se calculan las diferencias $c_j - z_j$.

$$c_2 - z_2 = 1 - (-2) = 3 \Rightarrow \text{entra } A_2$$

$$c_3 - z_3 = 0 - 2 = -2$$

5. Se determina el vector que sale.

$$\theta_2 = \min \left\{ \frac{x_1}{y_{12}}, \frac{x_4}{y_{42}} \right\} = \min \left\{ -; \frac{20}{1} \right\} = 20 \Rightarrow \text{sale } A_4$$

6. Se calcula la nueva solución, el z asociado y se vuelve al punto 2.

$$x'_1 = 10 - \theta_2 y_{12} = 10 - 20(-1) = 30; x'_2 = \theta_2 = 20; x'_3 = 0$$

$$x'_4 = 0;$$

$$x_B = (x_1, x_2)^T = (30; 20)^T; x_N = (x_3, x_4)^T = (0; 0)^T$$

$$z = c_B x_B = (2; 1)(30; 20)^T = 80$$

1. Se calculan los y_{ij} .

$$B^{-1} = \begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix}$$

$$Y = B^{-1}N = \begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} y_{13} & y_{14} \\ y_{23} & y_{24} \end{pmatrix}$$

2. Se determinan los z_j .

$$(z_3, z_4) = c_B Y = (2; 1) \begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix} = (-4; 3)$$

3. Se calculan las diferencias $c_j - z_j$.

$$c_3 - z_3 = 0 - (-4) = 4 \Rightarrow \text{entra } A_3$$

$$c_4 - z_4 = 0 - 3 = -3$$

4. Se determina el vector que sale.

$$\theta_1 = \min \left\{ \frac{x_1}{y_{13}}, \frac{x_2}{y_{23}} \right\} = \min \{-; -\} =$$

Aun no se ha llegado a la condición de óptimo pues

$C_3 - Z_3 > 0$, sin embargo

$y_{13} < 0$, por lo tanto A_1 no puede salir de la base y

$y_{23} < 0$, por lo tanto A_2 tampoco puede salir de la base.

La solución factible puede hacerse arbitrariamente grande.

El conjunto de soluciones factibles es no acotado en la dirección de C .